

$$1. \quad 1-1. \quad ① \quad |A - \lambda I| = \begin{vmatrix} 4-\lambda & 2 \\ 2 & 1-\lambda \end{vmatrix} = (4-\lambda)(1-\lambda) - 4 = 4 - 5\lambda + \lambda^2 - 4 = \lambda(\lambda - 5) = 0.$$

$$i) \lambda_1 = 0 \quad A\mathbf{x} = \mathbf{0} \quad \begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix} \mathbf{x} = \mathbf{0} \quad 4x + 2y = 0 \quad y = -2x. \quad w_1 = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

$$ii) \lambda_2 = 5 \quad (A - 5I)\mathbf{x} = \mathbf{0} \quad \begin{pmatrix} -1 & 2 \\ 2 & -4 \end{pmatrix} \mathbf{x} = \mathbf{0} \quad -x + 2y = 0 \quad x = 2y. \quad w_2 = \frac{1}{\sqrt{5}} \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$② \quad A = PDP^T. \quad P = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 & 2 \\ -2 & 1 \end{pmatrix} \quad D = \begin{pmatrix} 0 & 0 \\ 0 & 5 \end{pmatrix}$$

$$\Rightarrow A = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 & 2 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 5 \end{pmatrix} \frac{1}{\sqrt{5}} \begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix}$$

$$③ \quad \lambda_1 = 0, \lambda_2 = 5 \rightarrow \lambda_1, \lambda_2 \geq 0. \quad \therefore A; PSD$$

$$1-2. \quad ① \quad B^T B = \begin{pmatrix} 2 & 1 & 2 \\ 2 & -1 & 2 \end{pmatrix} \begin{pmatrix} 2 & 2 \\ 1 & -1 \\ 2 & 2 \end{pmatrix} = \begin{pmatrix} 4+1+4 & 4-1+4 \\ 4-1+4 & 4+1+4 \end{pmatrix} = \begin{pmatrix} 9 & 7 \\ 7 & 9 \end{pmatrix}$$

$$B B^T = \begin{pmatrix} 2 & 2 \\ 1 & -1 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} 2 & 1 & 2 \\ 2 & -1 & 2 \end{pmatrix} = \begin{pmatrix} 4+4 & 2-2 & 4+4 \\ 2-2 & 1+1 & 2-2 \\ 4+4 & 2-2 & 4+4 \end{pmatrix} = \begin{pmatrix} 8 & 0 & 8 \\ 0 & 2 & 0 \\ 8 & 0 & 8 \end{pmatrix}$$

$$② \quad |B^T B - \lambda I| = \begin{vmatrix} 9-\lambda & 7 \\ 7 & 9-\lambda \end{vmatrix} = (9-\lambda)^2 - 49 = 0. \quad 9-\lambda = \pm 7 \quad \lambda_1 = 2, \quad \lambda_2 = 16$$

$$i) \lambda_1 = 2 \quad \begin{pmatrix} 7 & 7 \\ 7 & 7 \end{pmatrix} \mathbf{x} = \mathbf{0} \quad x+y=0. \quad x=-y. \quad w_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$ii) \lambda_2 = 16 \quad \begin{pmatrix} -7 & 7 \\ 7 & -7 \end{pmatrix} \mathbf{x} = \mathbf{0} \quad -x+y=0. \quad x=y. \quad w_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\Rightarrow V = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}, \quad d_1 = \sqrt{2}, \quad d_2 = \sqrt{16} = 4$$

$$③ \quad i) \lambda_1 = 2 \quad B B^T - 2I = \begin{pmatrix} 6 & 0 & 8 \\ 0 & 0 & 0 \\ 8 & 0 & 6 \end{pmatrix} = \mathbf{0} \quad \begin{matrix} 6x+8z=0 \\ 8x+6z=0 \end{matrix} \quad \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad w_1 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$ii) \lambda_2 = 16 \quad B B^T - 16I = \begin{pmatrix} -8 & 0 & 8 \\ 0 & -14 & 0 \\ 8 & 0 & -8 \end{pmatrix} = \mathbf{0} \quad \begin{matrix} -8x+8z=0 \\ -14y=0 \\ 8x-8z=0 \end{matrix} \quad \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad w_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

$$iii) \lambda_3 = 0 \quad B B^T = \begin{pmatrix} 8 & 0 & 8 \\ 0 & 2 & 0 \\ 8 & 0 & 8 \end{pmatrix} = \mathbf{0}. \quad \begin{matrix} 8x+8z=0 \\ 2y=0 \end{matrix} \quad \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad w_3 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

$$\therefore U = \begin{pmatrix} 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}$$

$$④ \quad B = U \Sigma V^T = \begin{pmatrix} 0 & 1/\sqrt{2} & 1/\sqrt{2} \\ 1 & 0 & 0 \\ 0 & 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix} \begin{pmatrix} \sqrt{2} & 0 \\ 0 & 4 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix}$$

$$1-3. \quad ① \quad |\Sigma - \lambda I| = \begin{vmatrix} 5-\lambda & 3 \\ 3 & 5-\lambda \end{vmatrix} = (5-\lambda)^2 - 9 = 0. \quad 5-\lambda = \pm 3 \quad \lambda_1 = 2 \quad \lambda_2 = 8$$

$$i) \lambda_1 = 2 \quad \begin{pmatrix} 3 & 3 \\ 3 & 3 \end{pmatrix} \mathbf{x} = \mathbf{0} \quad x+y=0. \quad x=-y. \quad w_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$ii) \lambda_2 = 8 \quad \begin{pmatrix} -3 & 3 \\ 3 & -3 \end{pmatrix} \mathbf{x} = \mathbf{0} \quad -x+y=0 \quad x=y. \quad w_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$② \quad PC1 \text{ 방향성비율}; w_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \dots \quad \lambda_2 = 8 \quad (\text{가장 큰 값})$$

$$PC2 \text{ 방향성비율}; w_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad \dots \quad \lambda_1 = 2$$

$$③ \quad PC1 \text{ 분산 비율}; \frac{8}{10} = 0.8, \quad PC2 \text{ 분산 비율}; \frac{2}{10} = 0.2$$

$$④ \quad 1 - 0.8 = 0.2. \quad \underline{20\%} \quad \therefore \text{전체 분산 중 PC1이 } 80\% \text{를 설명하고 있으며, PC2가 설명하는 분산은 } 20\% \text{만큼 한정된다.}$$

$$2. \quad 2-1 \quad ① \quad P(A) = \frac{8}{16} = 0.5 \quad P(B) = \frac{4}{16} = 0.25 \quad P(C) = \frac{4}{16} = 0.25$$

$$② \quad H(x) = -\sum P(x_i) \log_2 P(x_i) = -[0.5 \log_2(0.5) + 0.25 \log_2(0.25) + 0.25 \log_2(0.25)]$$

$$= -[0.5 \cdot -1 + 0.25 \cdot -2 + 0.25 \cdot -2] = -(-0.5 - 0.5 - 0.5) = 1.5$$

$$③ \quad \text{콜라츠: } 2개 \rightarrow \text{두 콜라츠 확률} = \frac{1}{3} \text{ 일 때 발생.}$$

2-2. ① $D_{KL}(P||Q) = \sum P(x) \log_2 \frac{P(x)}{Q(x)} = 0.5 \log_2 \frac{0.5}{0.25} + 0.3 \log_2 \frac{0.3}{0.4} + 0.2 \log_2 \frac{0.2}{0.25} = 0.5 - 0.221 - 0.064 \approx 0.215$

$D_{KL}(Q||P) = \sum Q(x) \log_2 \frac{Q(x)}{P(x)} = 0.25 \log_2 \frac{0.25}{0.5} + 0.4 \log_2 \frac{0.4}{0.3} + 0.25 \log_2 \frac{0.25}{0.2} = -0.25 + 0.368 + 0.080 \approx 0.199$

$\therefore D_{KL}(P||Q) = 0.215 \neq D_{KL}(Q||P) = 0.199$ 대칭 $\times \rightarrow$ 거리의 조건에 만족하지 않는다.

② 실제 분포가 P인데, 이를 Q 분포로 근사했을 때에 발생하는 정보량 손실 정도. P 관점에서 Q, P 분포 차이.

$P(x) = Q(x)$ 일 때, $D_{KL}(P||Q) = 0$.

3. 3-1 ① $\bar{x} = \frac{1+2+3+4+5}{5} = \frac{15}{5} = 3, \quad \bar{y} = \frac{3+4+5+6+9}{5} = \frac{10+9}{5} = 5$

$S_{xy} = (1-3)(2-5) + (2-3)(3-5) + (3-3)(4-5) + (4-3)(6-5) + (5-3)(9-5) = 6 + 2 + 1 + 8 = 17$

$S_{yy} = 4 + 1 + 1 + 4 = 10$.

② $\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{17}{10} = 1.7, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 5 - 1.7 \cdot 3 = 5 - 5.1 = -0.1$

$\therefore \hat{y} = -0.1 + 1.7x$

③ $SSE = \sum e^2 = 0.16 + 0.09 + 0.49 + 0.36 = 1.10$

$SST = \sum (y_i - \bar{y})^2 = 9 + 4 + 1 + 16 = 30$.

$R^2 = 1 - \frac{SSE}{SST} = 1 - \frac{1.10}{30} = 1 - 0.0367 \approx 0.963$

④ $R^2 = 0.963$ 즉 약 96.3% 설명. 설명력 좋음.

3-2. ① $VIF = \frac{1}{1-R^2} \Rightarrow VIF_1 = \frac{1}{1-0.9} = \frac{1}{0.1} = 10$

② $VIF = 10$ 인지 다중공선성이 심각한 수준임 $\rightarrow X_1$ 을 모델에서 제거하거나 PCA 차원 축소, Ridge 기법을 사용해야 한다.

4. 4-1 ① Class A. $d(q, (1,1)) = \sqrt{9+9} = 3\sqrt{2} \approx 4.243$

$d(q, (3,4)) = \sqrt{0.25} = 0.5$

Class B $d(q, (4,4)) = \sqrt{1+1} = \sqrt{2} \approx 1.414$

$d(q, (5,3)) = \sqrt{1+1} = \sqrt{2}$

② $(3,4,4) = A, (4,4) = B, (5,3) = B$. 거리 이었 $\Rightarrow$ Class B 분류

③ $w_{(3,4,4)} = \frac{1}{0.5} = 2, \quad w_{(4,4)} = \frac{1}{1.414} \approx 0.707, \quad w_{(5,3)} = \frac{1}{1.414}$

$w_A = 2, \quad w_B = (0.707 \times 2) = 1.414, \quad \Rightarrow w_A > w_B, \quad \text{Class A 분류}$

④ 단순 다중점 KNN은 Class B, Weighted KNN은 Class B로 예측. 결과 다름.

단순 다중점 방법은 거리 정보를 반영하지 않음. 더 가까운 이웃일수록 더 큰 영향력 가짐.

5. 5-1 ① $P_1(2,7) \dots d(C_1) = 0, d(C_2) = \sqrt{36+5} = \sqrt{41} \approx 7.81 \Rightarrow C_1$

$P_2(2,1) \dots d(C_1) = 2, d(C_2) = \sqrt{36+49} = \sqrt{85} \approx 9.22 \Rightarrow C_1$

$P_3(7,2) \dots d(C_1) = \sqrt{2} = 1.41, d(C_2) = \sqrt{25+90} = \sqrt{115} \approx 10.71 \Rightarrow C_1$

$P_4(8,8) \dots d(C_1) = \sqrt{61} \approx 7.81, d(C_2) = 0 \Rightarrow C_2$

$P_5(9,8) \dots d(C_1) = \sqrt{49+9} = \sqrt{58} \approx 7.6, d(C_2) = 1 \Rightarrow C_2$

$P_6(8,9) \dots d(C_1) = \sqrt{72} \approx 8.49, d(C_2) = 1 \Rightarrow C_2$

$\therefore C_1: P_1, P_2, P_3, C_2: P_4, P_5, P_6$

② $C_1 = \left(\frac{2+2+7}{3}, \frac{7+1+2}{3} \right) = \left(\frac{11}{3}, 2 \right) = (2.777, 2)$

$C_2 = \left(\frac{8+8+9}{3}, \frac{8+8+9}{3} \right) = \left(\frac{25}{3}, \frac{25}{3} \right) = (8.333, 8.333)$

③ $P_1(2,7) \dots d(C_1) = \sqrt{0.777^2+1} \approx 1.05, d(C_2) = \sqrt{6.777^2+5.777^2} \approx 8.28 \Rightarrow C_1$

$P_2(2,1) \dots d(C_1) = \sqrt{0.777^2+1} \approx 1.05, d(C_2) = \sqrt{6.777^2+7.777^2} \approx 9.69 \Rightarrow C_1$

$P_3(7,2) \dots d(C_1) = \sqrt{0.667^2} = 0.667, d(C_2) = \sqrt{5.777^2+6.777^2} \approx 8.28 \Rightarrow C_1$

$P_4(8,8) \dots d(C_1) = \sqrt{5.667^2+6^2} \approx 8.25, d(C_2) = \sqrt{0.777^2+0.777^2} \approx 0.47 \Rightarrow C_2$

$P_5(9,8) \dots d(C_1) = \sqrt{6.667^2+6^2} \approx 8.97, d(C_2) = \sqrt{0.667^2+0.777^2} \approx 0.95 \Rightarrow C_2$

$P_6(8,9) \dots d(C_1) = \sqrt{5.667^2+7^2} \approx 9.01, d(C_2) = \sqrt{0.777^2+0.667^2} \approx 0.97 \Rightarrow C_2$

$\therefore C_1: P_1, P_2, P_3, C_2: P_4, P_5, P_6 \rightarrow$ 병합됨.

$\Rightarrow$ 병합 결과가 변하지 않아, 알고리즘 2회 반복은 수렴.

5-2 ① $d(P_1, P_2) = 1, d(P_1, P_3) = 1, d(P_1, P_4) = \sqrt{2}, d(P_2, P_3) = \sqrt{2}, d(P_2, P_4) = 1, d(P_3, P_4) = 1$ 모든 $\leq 1.5 \rightarrow$ 전방서로 이동

$d(P_5, P_6) = 1, d(P_5, P_7) = 2, d(P_5, P_8) = 1$

$d(P_6, P_7) = \sqrt{2}, d(P_6, P_8) = \sqrt{2}, d(P_7, P_8) = 1, P_9$; 다른 모든 점과 거리 4 이상

$\Rightarrow P_1, P_2, P_3, P_4, P_5, P_6, P_7, P_8$; 4개 P_5, P_6 ; 2개 P_7 ; 2개 P_9 ; 1개

② $\min P_k = 3 \rightarrow$ 이웃 개수 $\geq 3 \Rightarrow$ core point.

$P_1, P_2, P_3, P_4, P_5, P_6, P_8$: Core point P_7 : Border P_9 : Noise

③ DBSCAN은 K-means와 달리 중심점 미리 정하는 방식이 아닌, 밀도 기준으로 점 연결하는 방식.

전체 데이터 순회하며 core point와의 연접 관계로 군집 형성, $\rightarrow$ 이웃을 진행하며 군집 개수 & 데이터 분포 따라 결정됨.